SMITH ENGINEERING

Queen's University

Department of Electrical and Computer Engineering



ELEC 490/498 Project Blueprint

Autonomous Smart Caddy Cart (ASCC)

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Submitted by Group 07

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Acronyms

1. Introduction

1.1 Motivation and Problem

Golfers who walk the course must transport their equipment by carrying a bag or operating a push cart. Although electric and autonomous carts reduce this effort, they primarily focus on transportation rather than equipment management. Golfers must also keep track of clubs removed during play, which can be misplaced or left behind.

The Autonomous Smart Caddy Cart (ASCC) aims to address both challenges by combining autonomous golfer following with club identification and tracking. As golfers ourselves, the team understands the need for a solution that addresses both transportation and equipment management. The goal is to reduce manual cart operation while helping golfers manage their equipment throughout a round.

1.2 Project Objectives

The ASCC will be developed as a motorized golf push cart capable of following a golfer using a controller-based tracking system rather than computer vision. Distance sensors will support obstacle detection and avoidance, while an emergency-stop mechanism will allow the cart to be stopped when necessary.

An RFID-based subsystem will identify tagged golf clubs and monitor their presence in the bag, notifying the golfer when a club is missing. The objective is to integrate these functions into a reliable and safe prototype.

The ASCC will also aim to incorporate golf course data, including hole layouts, distances, and course information, to assist golfers with shot planning. This information helps golfers estimate distances to targets and hazards, supporting more informed club selection and on-course decisions.

1.3 Project Scope

The minimum viable prototype will focus on motor control, golfer following, obstacle detection, emergency stopping, RFID-based club tracking, and basic golf course data integration. Additional features, including shot data collection, club recommendations, and speech functionalities, may be implemented if time and resources permit.

Testing will begin indoors before progressing to controlled outdoor environments. The prototype is not intended for commercial deployment or operation on public roads, crowded areas, or extreme terrain.

1.4 System Overview and Operating Scenario

Figure 1 illustrates the ASCC's main subsystems and their interactions with the golfer and surrounding environment. The Raspberry Pi coordinates sensor inputs, motor control, and club tracking.

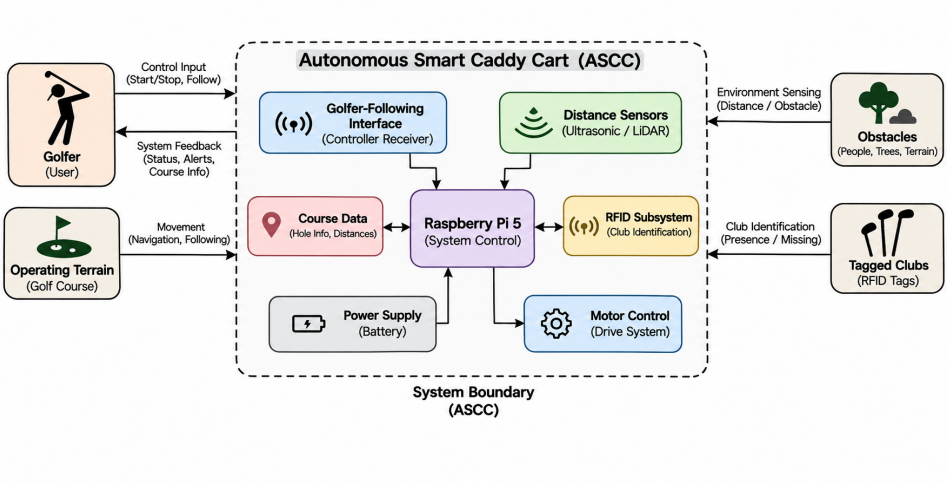


Figure 1: ASCC system overview and external interactions.

1.5 Design Constraints

The ASCC must be completed within the CAD $600 budget and capstone development schedule. Existing components, including the Raspberry Pi, ultrasonic sensors, and golf push cart, will be reused to reduce costs.

Key constraints include hardware compatibility, sensor reliability, battery capacity, and safe motor operation. These considerations will guide the system requirements, development approach, and verification procedures.

2. Functional Description

2.1 System Overview and Design Alternatives

2.2 Functional Decomposition

2.3 Logical Decomposition and Interfaces

2.4 Physical Architecture

2.5 Prototype Definitions

3. Requirements Specifications

3.1 Stakeholder Needs and Requirements Hierarchy

The ASCC requirements originate from five stakeholder needs: equipment transportation (SN-01), safety (SN-02), club management (SN-03), course information (SN-04), and reliable operation (SN-05). These are decomposed into system-level (SYS) and subsystem requirements covering golfer following (FL), obstacle detection (OB), motor control (MC), club tracking (CT), course data (CD), and power management (PW). This hierarchy links stakeholder needs to verifiable specifications and tests. Tables 2 and 3 detail the hierarchy.

3.2 Functional Requirements

Functional requirements define the ASCC's core operations: golfer following, obstacle detection, emergency stopping, RFID club identification, and course information display. Broad functions are decomposed into independently verifiable "shall" statements. For example, golfer following requires interpreting tracking information and updating motor commands, while club tracking requires RFID identification and missing-club alerts. Table 4 lists the functional requirements.

3.3 Non-Functional and Interface Requirements

Performance targets include a 1–3 m following distance, 1.5 m/s walking speed, 1 m obstacle detection range, 90% RFID identification accuracy, and 30-minute battery runtime. These preliminary values balance user accessibility, walking-speed operation, and prototype feasibility. The detection range must be validated against stopping distance and sensor uncertainty, while RFID accuracy and runtime will be assessed experimentally. Table 5 details these specifications.

Interface requirements define the exchange of tracking data, sensor measurements, motor commands, RFID identifiers, user feedback, and electrical power. These correspond to the labelled information and energy flows in the functional decomposition and are listed in Table 6.

3.4 Safety and Fault Requirements

Safety requirements use explicit “shall not” statements to prohibit unsafe movement during emergency-stop activation, tracking-signal loss, obstacle proximity, or invalid sensor readings. The proposed 1 s tracking timeout and obstacle stopping zone will be validated through controlled fault-injection and braking tests. Emergency stopping takes priority over autonomous commands and requires manual reset. Tables 7 and 10 define these requirements and their verification procedures.

3.5 Critical Requirements and Traceability

Requirements are classified as critical (safety and loss of control), major (essential functionality), or minor (supporting features). Each requirement traces to a stakeholder need, parent requirement, subsystem, prototype, and V&V test.

For example, critical requirement MC-04 prohibits motor actuation while the emergency stop is engaged. Derived from SN-02 and SYS-03, it is allocated to the motor-control safety interlock and verified through VT-07. Similar relationships are established for tracking loss (FL-04) and obstacle faults (OB-04, OB-05). Tables 8, 9, and 11 provide detailed traceability.

3.6 Stretch Goals

Optional features include shot data collection (SG-01), club recommendations (SG-02), and speech interaction (SG-03). These use “may” statements and are excluded from mandatory prototype acceptance criteria. Table 12 lists the stretch goals.

4. Risks and Safety Hazards

4.1 Hazard Identification and Risk Assessment

4.2 Risk Mitigation and Residual Risk

5. Approach

5.1 Development Strategy and Division of Labour

5.2 Prototype Development Plan

5.3 Milestones and Dependencies

5.4 Contingency Plans

6. Verification and Validation

6.1 Verification Strategy and Traceability

6.2 Subsystem and Prototype Tests

6.3 Experimental Test Setups

6.4 System Validation

7. Resource Requirements

The ASCC has a project budget of CAD $600. Several key components, including the Raspberry Pi, ultrasonic sensors, and golf push cart, are already owned or borrowed, reducing the overall cost. The remaining budget provides flexibility for unexpected expenses, replacements, or design changes. Additional components, including motors and GPS tracking hardware, may be obtained through the ECE Department or Ingenuity Labs. However, their estimated costs are included in the budget until their availability is confirmed. Table 1 summarizes the estimated costs.

7.1 Hardware and Equipment

Table 1: Estimated ASCC bill of materials (CAD).

Component	Qty.	Cost
Raspberry Pi 5 (existing)	1	$0
Raspberry Pi display	1	$40
Ultrasonic sensors (existing)	Multiple	$0
RFID readers and tags	2 sets	$20
TF-Luna LiDAR sensor	1	$50
GPS tracker	1	$30
Golf push cart (existing)	1	$0
Motors	1 set	$80
Motor controller	1	$50
Battery and power supply	1 set	$75
Mounting hardware	Assorted	$30
Wiring, connectors, switches	Assorted	$30
Tax and shipping	–	$75
Total		**$480**

7.2 Facilities and Contingencies

Development will require electronics tools and access to indoor and outdoor testing areas. Tools can be found utilizing the ’65 ECE lab, and testing will occur in controlled areas and hopefully on a real golf course for the finished prototype. Python and Git will support software development.

The estimated cost is $480, leaving $120 below the $600 budget limit. This remaining budget allows for additional components or design adjustments if needed.

8. Conclusion

The ASCC blueprint builds upon the initial proposal by establishing a more detailed system design, measurable performance requirements, and contingency plans for potential failures. The project continues to focus on autonomous golfer following, obstacle de-

tection, emergency stopping, RFID-based club tracking, and golf course information to improve the walking golf experience.

A key focus of the blueprint is justifying quantitative performance targets based on hardware capabilities, operating conditions, and safety considerations. These targets provide measurable acceptance criteria for verification and validation testing, ensuring that system performance can be evaluated against clearly defined requirements. The incremental development approach also allows individual prototypes to be tested before integration, reducing the risk of unresolved subsystem failures.

Potential limitations include unreliable sensor readings, inconsistent RFID detection, loss of golfer tracking, and difficulties operating on uneven terrain. The blueprint addresses these risks through alternative hardware configurations, adjustments to sensor placement, controlled testing, and fail-safe stopping behaviour. If technical or scheduling constraints arise, optional features will be deferred to prioritize safe and reliable operation of the core system.

With an estimated budget of CAD $480 and access to existing equipment, the project remains feasible within the $600 limit. The team has established its system architecture and identified the major technical requirements and resources. The next steps involve finalizing quantitative acceptance criteria, confirming hardware availability, and developing subsystem prototypes to validate the proposed design before full integration.

9. References

A. Detailed Requirements and Traceability

This appendix provides the complete ASCC requirements hierarchy, subsystem specifications, and verification traceability. Requirements are organized from stakeholder needs (SN) to system-level requirements (SYS) and subsystem requirements. Each requirement is assigned a priority and linked to a planned verification test.

A.1 Stakeholder Needs and System Requirements

Table 2: Stakeholder needs and associated system requirements.

ID	Stakeholder Need	System Requirements
SN-01	Transport golf equipment without manually pushing the cart.	SYS-01
SN-02	Prevent collisions and unintended movement.	SYS-02, SYS-03
SN-03	Identify clubs and notify users of missing equipment.	SYS-04
SN-04	Access course information and distances.	SYS-05
SN-05	Maintain reliable operation within project constraints.	SYS-06

Table 3: System-level requirements and classifications.

ID	System Requirement	Priority
SYS-01	The ASCC shall autonomously follow a golfer carrying the designated tracking controller.	Major
SYS-02	The ASCC shall prevent forward movement into detected obstacles within its stopping zone.	Critical
SYS-03	The ASCC shall provide an emergency-stop function that overrides autonomous movement commands.	Critical
SYS-04	The ASCC shall monitor the presence of individually tagged golf clubs.	Major
SYS-05	The ASCC shall provide available golf course information through a user interface.	Minor
SYS-06	The ASCC shall support battery-powered operation during controlled testing.	Major

A.2 Functional Subsystem Requirements

Table 4: Detailed functional subsystem requirements.

ID	Requirement	Parent
FL-01	The following subsystem shall calculate the required movement direction from valid tracking-controller information.	SYS-01
FL-02	The following subsystem shall update its movement command when the golfer changes position relative to the cart.	SYS-01
OB-01	The obstacle subsystem shall measure distances to objects in the forward detection region.	SYS-02
OB-02	The obstacle subsystem shall issue a stop request when a detected obstacle enters the stopping zone.	SYS-02
MC-01	The motor-control subsystem shall execute valid movement commands received from the main controller.	SYS-01
MC-02	The motor-control subsystem shall disable motor actuation in response to an emergency-stop request.	SYS-03
CT-01	The club-tracking subsystem shall associate detected RFID identifiers with registered golf clubs.	SYS-04
CT-02	The club-tracking subsystem shall generate a missing-club alert when a registered club remains absent beyond the configured detection interval.	SYS-04
CD-01	The course-data subsystem shall display available hole information for the selected course.	SYS-05
CD-02	The course-data subsystem shall display an estimated distance to a selected course target when valid location data is available.	SYS-05

A.3 Performance and Operational Requirements

Table 5: Detailed performance and operational requirements.

ID	Requirement	Parent
FL-03	The ASCC shall maintain a following distance between 1 and 3 m during controlled walking trials.	SYS-01
MC-03	The ASCC shall achieve a forward speed of at least 1.5 m/s on level terrain under the specified test load.	SYS-01
OB-03	The obstacle subsystem shall detect the designated test obstacle at a distance of at least 1 m.	SYS-02
CT-03	The RFID subsystem shall correctly identify at least 90% of registered clubs during repeated detection trials.	SYS-04
PW-01	The power subsystem shall support at least 30 minutes of continuous operation under representative test conditions.	SYS-06

The following distance of 1–3 m provides an initial separation target while keeping equipment accessible. The 1.5 m/s speed requirement approximates walking speed. The obstacle detection distance will be evaluated against measured stopping distance and control latency.

The RFID accuracy target provides a measurable baseline for evaluating reader placement, while the runtime requirement supports repeated prototype demonstrations. These are preliminary targets requiring validation through hardware testing and supporting calculations.

A.4 Interface Requirements

Interface requirements define the exchange of information, commands, and electrical power between logical subsystems.

Table 6: Detailed interface requirements.

ID	Requirement	Parent
IR-01	The following interface shall transmit tracking information to the main controller.	SYS-01
IR-02	The obstacle sensors shall transmit distance measurements to the main controller.	SYS-02
IR-03	The main controller shall transmit movement commands to the motor-control subsystem.	SYS-01
IR-04	The RFID interface shall transmit detected tag identifiers to the main controller.	SYS-04
IR-05	The user interface shall display club-status information generated by the club-tracking subsystem.	SYS-04
IR-06	The power subsystem shall provide the required operating voltages to the computing and motor-control electronics.	SYS-06

A.5 Safety and Fault Requirements

Safety requirements define prohibited operating conditions and expected responses to faults. These requirements must be verified before integrated outdoor testing.

Table 7: Detailed safety and fault requirements.

ID	Requirement	Parent
MC-04	The ASCC shall not energize its drive motors while the emergency-stop control is engaged.	SYS-03
MC-05	The ASCC shall not resume autonomous movement following an emergency stop until a manual reset is performed.	SYS-03
FL-04	The ASCC shall not continue autonomous movement when the tracking signal has been unavailable for more than 1 s.	SYS-03
OB-04	The ASCC shall not continue forward movement when an obstacle is within the validated stopping zone.	SYS-02
OB-05	The ASCC shall not initiate autonomous movement when required obstacle sensor data is unavailable or invalid.	SYS-02

The proposed 1 s tracking timeout is an initial safety target. The stopping-zone threshold will be determined from measured response latency, braking distance, and sensor uncertainty. Invalid sensor data and loss of tracking will result in a stopped state rather than continued autonomous movement.

A.6 Complete Requirements Traceability Matrix

The traceability matrix links each subsystem requirement to its parent system requirement, stakeholder need, assigned prototype, priority, and planned verification test.

Table 8: Functional and performance requirements traceability.

ID	Parent	Need	Priority	Prototype	Test
FL-01	SYS-01	SN-01	Major	P-02	VT-02
FL-02	SYS-01	SN-01	Major	P-02	VT-02
FL-03	SYS-01	SN-01	Major	P-02	VT-02
FL-04	SYS-03	SN-02	Critical	P-02	VT-08
OB-01	SYS-02	SN-02	Critical	P-03	VT-03
OB-02	SYS-02	SN-02	Critical	P-03	VT-04
OB-03	SYS-02	SN-02	Critical	P-03	VT-03
OB-04	SYS-02	SN-02	Critical	P-03	VT-04
OB-05	SYS-02	SN-02	Critical	P-03	VT-11
MC-01	SYS-01	SN-01	Major	P-01	VT-01
MC-02	SYS-03	SN-02	Critical	P-01	VT-07
MC-03	SYS-01	SN-01	Major	P-01	VT-01
MC-04	SYS-03	SN-02	Critical	P-01	VT-07
MC-05	SYS-03	SN-02	Critical	P-01	VT-07
CT-01	SYS-04	SN-03	Major	P-04	VT-05
CT-02	SYS-04	SN-03	Major	P-04	VT-05
CT-03	SYS-04	SN-03	Major	P-04	VT-05
CD-01	SYS-05	SN-04	Minor	P-05	VT-06
CD-02	SYS-05	SN-04	Minor	P-05	VT-06
PW-01	SYS-06	SN-05	Major	P-06	VT-09

Table 9: Interface requirements traceability.

ID	Parent	Need	Priority	Prototype	Test
IR-01	SYS-01	SN-01	Major	P-02	VT-10
IR-02	SYS-02	SN-02	Critical	P-03	VT-10
IR-03	SYS-01	SN-01	Major	P-01	VT-10
IR-04	SYS-04	SN-03	Major	P-04	VT-10
IR-05	SYS-04	SN-03	Major	P-05	VT-10
IR-06	SYS-06	SN-05	Major	P-06	VT-10

A.7 Verification Test Definitions

Table 10: Planned verification tests.

ID	Test	Verification Objective
VT-01	Motor movement	Verify movement commands and target speed.
VT-02	Golfer following	Measure following distance and response to golfer movement.
VT-03	Obstacle detection	Measure detection range and sensor response.
VT-04	Obstacle response	Verify stopping behaviour when obstacles enter the stopping zone.
VT-05	RFID identification	Measure identification accuracy and missing-club alerts.
VT-06	Course information	Verify hole information and displayed distances.
VT-07	Emergency stop	Verify motor inhibition and manual-reset behaviour.
VT-08	Signal loss	Verify stopping behaviour following tracking-signal loss.
VT-09	Battery runtime	Measure continuous operating duration.
VT-10	Interface integration	Verify information and power exchange between subsystems.
VT-11	Sensor fault	Verify safe response to missing or invalid obstacle measurements.

A.8 Critical Requirement Relationships

Critical requirements are linked to stakeholder needs, parent requirements, functions, logical elements, physical components, and verification tests.

Table 11: Critical requirement relationships.

ID	Function	Logical Element	Verification
MC-04	Inhibit motor actuation	Safety interlock and motor-control subsystem	VT-07
FL-04	Detect loss of tracking	Following controller and signal monitor	VT-08
OB-04	Prevent forward movement near obstacles	Obstacle monitor and safety controller	VT-04
OB-05	Detect invalid sensor information	Sensor validation logic	VT-11

For MC-04, the emergency-stop control must prevent motor actuation independently of normal movement commands. This requirement is derived from SYS-03 and SN-02. Verification will involve activating the emergency stop during controlled motor operation and confirming that motor actuation is disabled.

For FL-04, loss of the tracking signal must cause the cart to stop rather than continue using outdated commands. For OB-04 and OB-05, the obstacle-detection subsystem must prevent movement when an obstacle enters the stopping zone or required sensor information becomes invalid.

A.9 Stretch Goal Requirements

Stretch goals are optional and are not included in the minimum viable prototype acceptance criteria.

Table 12: Optional stretch goal requirements.

ID	Optional Requirement
SG-01	The ASCC may record shot locations and distances for later review.
SG-02	The ASCC may recommend golf clubs using course information and recorded shot data.
SG-03	The ASCC may provide speech-based status updates and user interaction.